

Attention mechanisms

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CAS Building ML/AI Applications, ETH Zurich, weekend 2

Friday 18 September 2026



Section 1

Attention

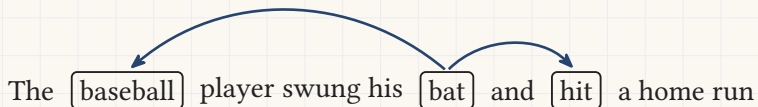
Attention

Words can find their meaning in a sentence by **attending to** other words!

The baseball player swung his bat and hit a home run

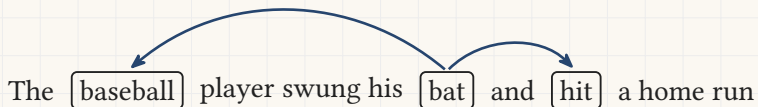
Attention

Words can find their meaning in a sentence by **attending to** other words!



Attention

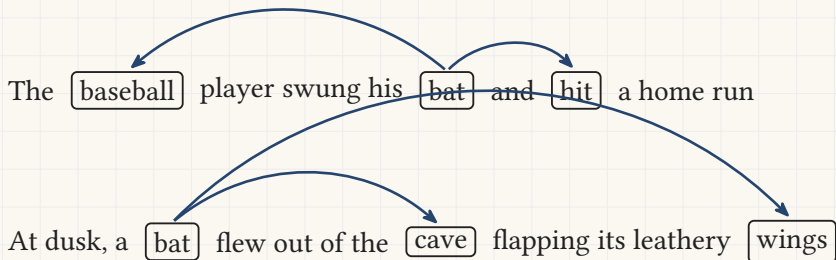
Words can find their meaning in a sentence by **attending to** other words!



At dusk, a bat flew out of the cave flapping its leathery wings

Attention

Words can find their meaning in a sentence by **attending to** other words!



Attention as information retrieval

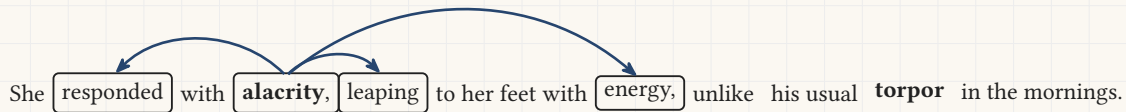
- She responded with **alacrity**, leaping to her feet with energy, unlike his usual **torpor** in the mornings.
- What do the words “alacrity” and “torpor” mean?

Attention as information retrieval



- She responded with **alacrity**, leaping to her feet with energy, unlike his usual **torpor** in the mornings.
- What do the words “alacrity” and “torpor” mean?
- Alacrity: brisk and cheerful readiness.
- Torpor: a state of physical or mental inactivity; lethargy.

Attention as information retrieval



Attention as information retrieval

She responded with **alacrity**, leaping to her feet with energy, **unlike** his usual **torpor** in the mornings.

The diagram illustrates the concept of attention as information retrieval. It features a sentence with three words highlighted in boxes: 'alacrity', 'unlike', and 'torpor'. A long blue curved arrow originates from the 'unlike' box and points to the 'alacrity' box, indicating a retrieval of information from a distant part of the sentence. A shorter blue curved arrow originates from the 'unlike' box and points to the 'torpor' box, indicating a retrieval of information from a closer part of the sentence.

Defining attention mechanisms

- We **define** a **value** representation for each word.

Defining attention mechanisms

- We **define** a **value** representation for each word.
- We let other words **enrich** their representations by **attending** to other words.

Creating embeddings for words

- Recall the Wuggish game. Every sentence describes at most three things.

Creating embeddings for words

- Recall the Wuggish game. Every sentence describes at most three things.



streept kat



bootet kat

Creating embeddings for words

- Recall the Wuggish game. Every sentence describes at most three things.



streept kat



vennomus snaak wudden chaar



bootet kat



streept snaak streept kat wudden chaar

Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?

Creating representations

- There are 8 words.
- There are 124 possible sentences.
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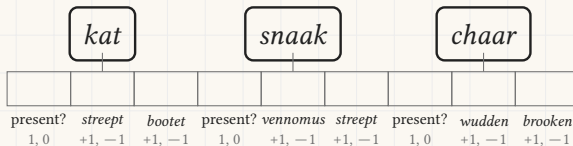
vennomus streept snaak wudden chaar



kat brooken chaar

Creating representations

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vennomus streept snaak wudden chaar



kat brooken chaar

Creating representations

- There are 8 words.
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<i>kat</i>			<i>snaak</i>			<i>chaar</i>		
present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>
1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1
0	0	0	1	+1	+1	1	+1	-1



vennomus streept snaak wudden chaar



kat brooken chaar

Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?



vennomus streept snaak wudden chaar

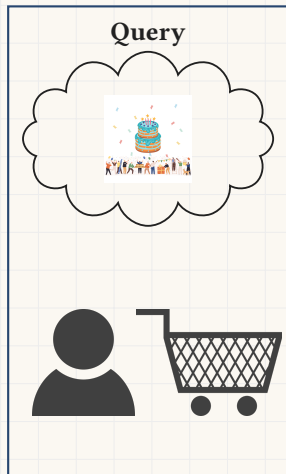


kat brooken chaar

	<i>kat</i>			<i>snaak</i>			<i>chaar</i>	
present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>
1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1

A	0	0	0	1	+1	+1	1	+1	-1
B	1	-1	-1	0	0	0	1	-1	+1

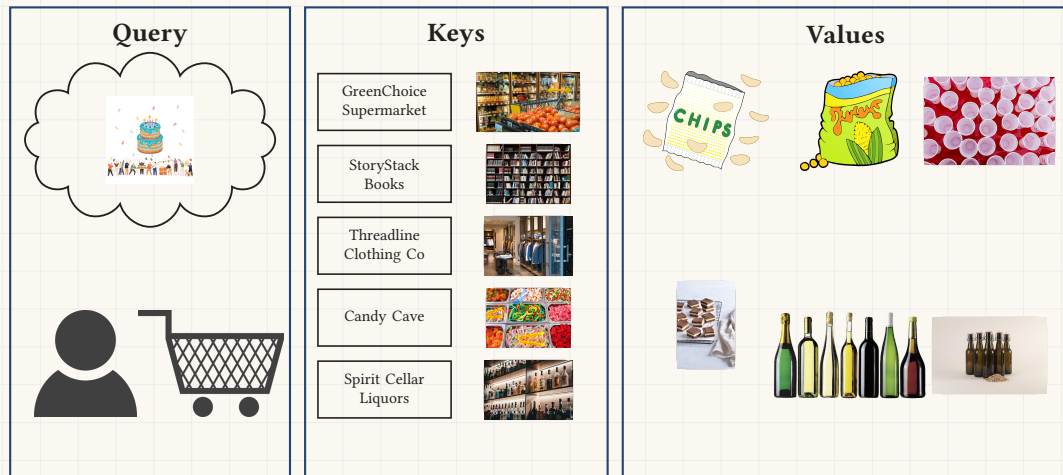
Attention as information retrieval



Attention as information retrieval



Attention as information retrieval



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Section 2

The inner product

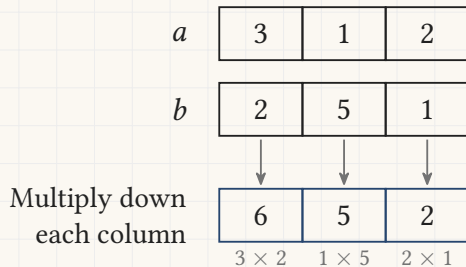
The inner product

$$a \quad \begin{array}{|c|c|c|} \hline 3 & 1 & 2 \\ \hline \end{array}$$

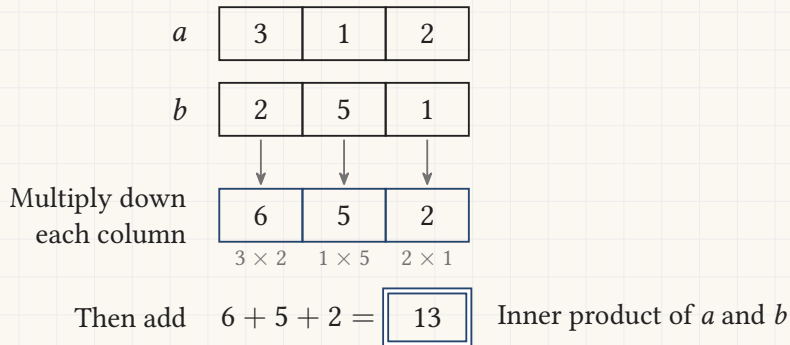
The inner product

a	3	1	2
b	2	5	1

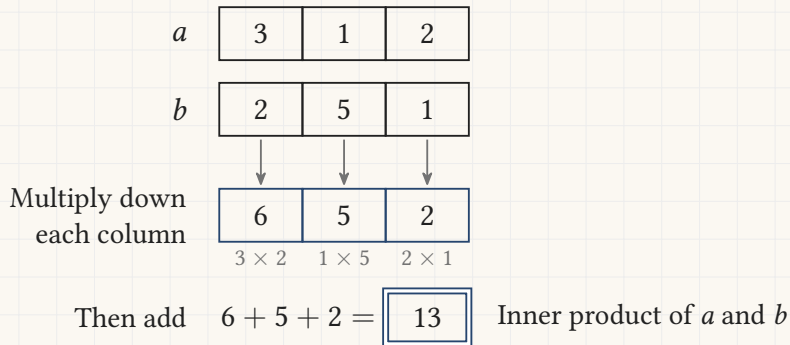
The inner product



The inner product



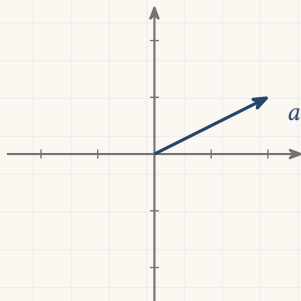
The inner product



$$\text{Inner product of } a \text{ and } b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

The inner product measures similarity

$$a = (2, 1)$$



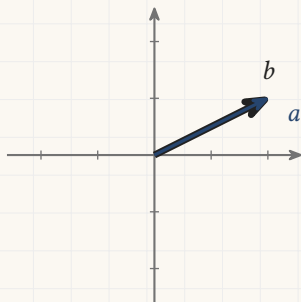
The inner product measures similarity

$$a = (2, 1)$$

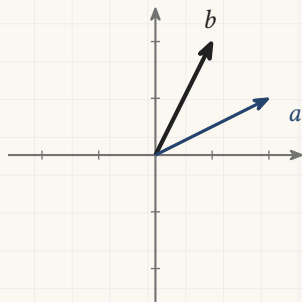
$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

Same direction



The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

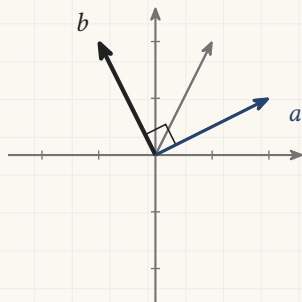
Same direction

$$b = (1, 2)$$

$$2 \times 1 + 1 \times 2 = 4$$

Close directions

The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

Same direction

$$b = (1, 2)$$

$$2 \times 1 + 1 \times 2 = 4$$

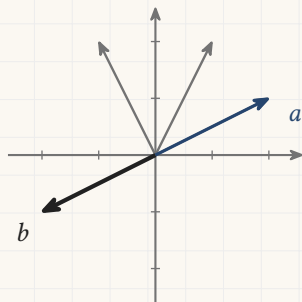
Close directions

$$b = (-1, 2)$$

$$2 \times (-1) + 1 \times 2 = 0$$

At right angles

The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1) \quad 2 \times 2 + 1 \times 1 = 5$$

Same direction

$$b = (1, 2) \quad 2 \times 1 + 1 \times 2 = 4$$

Close directions

$$b = (-1, 2) \quad 2 \times (-1) + 1 \times 2 = 0$$

At right angles

$$b = (-2, -1) \quad 2 \times (-2) + 1 \times (-1) = -5$$

Opposite directions

The inner product of zeros and ones

Query of *kat*

<i>kat</i>	<i>snaak</i>	<i>chaar</i>
1	0	0

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1
Key of <i>wudden</i>	0	0	1	0

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1
Key of <i>wudden</i>	0	0	1	0
Key of <i>streept</i>	1	1	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	
Query of <i>kat</i>	1	0	0		
Key of <i>kat</i>	1	0	0	1	It counts the slots where both have a 1
Key of <i>bootet</i>	1	0	0	1	
Key of <i>wudden</i>	0	0	1	0	
Key of <i>streept</i>	1	1	0	1	

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Section 3

The attention mechanism

An attention map

wudden

kat

vennomus

chaar

bootet

snaak

brooken

An attention map

Queries

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
<i>wudden</i>	0	0	0
<i>kat</i>	1	0	0
<i>vennomus</i>	0	0	0
<i>chaar</i>	0	0	1
<i>bootet</i>	0	0	0
<i>snaak</i>	0	1	0
<i>brooken</i>	0	0	0

An attention map

	Queries			Keys		
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
<i>wudden</i>	0	0	0	0	0	1
<i>kat</i>	1	0	0	1	0	0
<i>vennomus</i>	0	0	0	0	1	0
<i>chaar</i>	0	0	1	0	0	1
<i>bootet</i>	0	0	0	1	0	0
<i>snaak</i>	0	1	0	0	1	0
<i>brooken</i>	0	0	0	0	0	1

An attention map

	Queries			Keys			Values								
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>			<i>snaak</i>			<i>chaar</i>		
							<i>present?</i>	<i>streept</i>	<i>bootet</i>	<i>present?</i>	<i>vennomus</i>	<i>streept</i>	<i>present?</i>	<i>wudden</i>	<i>brooken</i>
<i>wudden</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	+2	0
<i>kat</i>	1	0	0	1	0	0	1	-1	-1	0	0	0	0	0	0
<i>vennomus</i>	0	0	0	0	1	0	0	0	0	0	+2	0	0	0	0
<i>chaar</i>	0	0	1	0	0	1	0	0	0	0	0	0	1	-1	-1
<i>bootet</i>	0	0	0	1	0	0	0	0	+2	0	0	0	0	0	0
<i>snaak</i>	0	1	0	0	1	0	0	0	0	1	-1	-1	0	0	0
<i>brooken</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	+2

An attention map

	Queries			Keys			Values									Embeddings																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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An attention map

	Queries			Keys			Values									Embeddings								
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>			<i>snaak</i>			<i>chaar</i>			<i>kat</i>			<i>snaak</i>			<i>chaar</i>		
							present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>	present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>
<i>wudden</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	+2	0									
<i>kat</i>	1	0	0	1	0	0	1	-1	-1	0	0	0	0	0	0									
<i>vennomus</i>	0	0	0	0	1	0	0	0	0	0	+2	0	0	0	0									
<i>chaar</i>	0	0	1	0	0	1	0	0	0	0	0	0	1	-1	-1									
<i>bootet</i>	0	0	0	1	0	0	0	0	+2	0	0	0	0	0	0									
<i>snaak</i>	0	1	0	0	1	0	0	0	0	1	-1	-1	0	0	0									
<i>brooken</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	+2									

An attention map

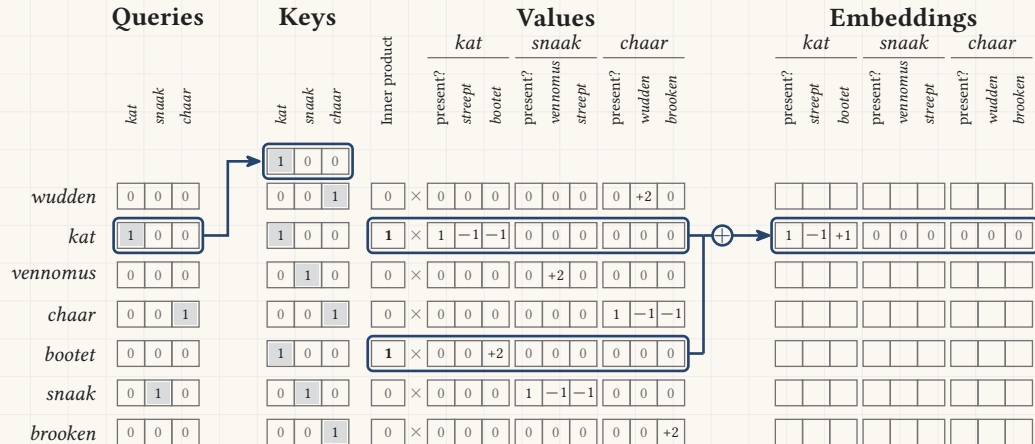
Queries			Keys			Values			Embeddings		
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>		<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
									<i>present?</i>	<i>streept</i>	<i>bootet</i>
<i>wudden</i>	0	0	0		0	0	1	0 × 0 0 0	0	0	0
<i>kat</i>	1	0	0		1	0	0	1 × 1 -1 -1	0	0	0
<i>vennomus</i>	0	0	0		0	1	0	0 × 0 0 0	0	+2	0
<i>chaar</i>	0	0	1		0	0	1	0 × 0 0 0	0	0	0
<i>bootet</i>	0	0	0		1	0	0	1 × 0 0 +2	0	0	0
<i>snaak</i>	0	1	0		0	1	0	0 × 0 0 0	1	-1	-1
<i>brooken</i>	0	0	0		0	0	1	0 × 0 0 0	0	0	+2

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
	<i>present?</i>	<i>streept</i>	<i>bootet</i>
<i>wudden</i>			
<i>kat</i>			
<i>vennomus</i>			
<i>chaar</i>			
<i>bootet</i>			
<i>snaak</i>			
<i>brooken</i>			

An attention map

Queries			Keys			Values									Embeddings												
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>		<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	<i>kat</i>			<i>snaak</i>			<i>chaar</i>			<i>kat</i>			<i>snaak</i>			<i>chaar</i>			
									present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>	present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>	
<i>wudden</i>	0	0	0		0	0	1	0 × 0 0 0	0	0	0	0	0	0	0	0	+2	0									
<i>kat</i>	1	0	0		1	0	0	1 × 1 -1 -1	0	0	0	0	0	0	0	0	0	0									
<i>vennomus</i>	0	0	0		0	1	0	0 × 0 0 0	0	+2	0	0	0	0	0	0	0	0									
<i>chaar</i>	0	0	1		0	0	1	0 × 0 0 0	0	0	0	0	0	0	1	-1	-1										
<i>bootet</i>	0	0	0		1	0	0	1 × 0 0 +2	0	0	0	0	0	0	0	0	0										
<i>snaak</i>	0	1	0		0	1	0	0 × 0 0 0	1	-1	-1	0	0	0	0	0	0										
<i>brooken</i>	0	0	0		0	0	1	0 × 0 0 0	0	0	0	0	0	0	0	0	+2										

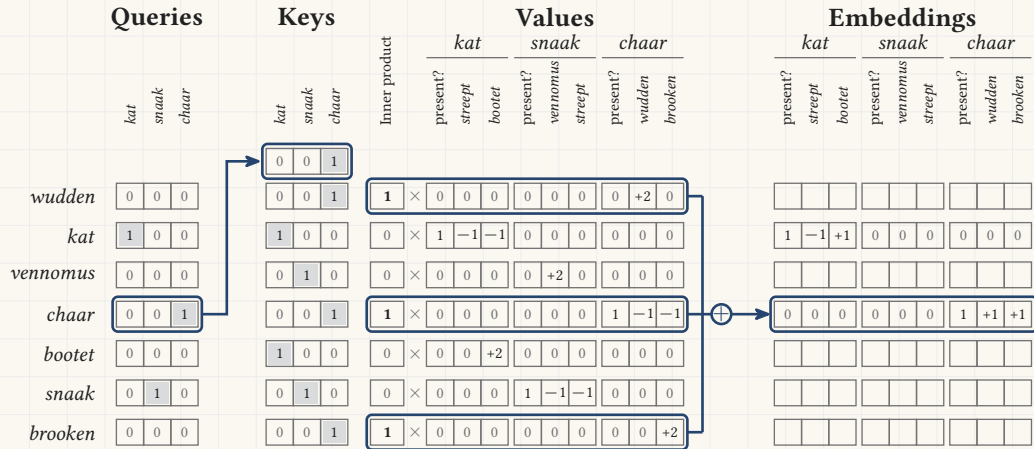
An attention map



An attention map

	Queries			Keys			Inner product	Values									Embeddings								
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>		<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>						
								present?	streept	bootet	present?	vennomus	streept	present?	wudden	brooken	present?	streept	bootet	present?	vennomus	streept	present?	wudden	brooken
<i>wudden</i>	0	0	0	0	0	1	1	×	0	0	0	0	0	0	0	0	+2	0							
<i>kat</i>	1	0	0	1	0	0	0	×	1	-1	-1	0	0	0	0	0	0	0	1	-1	+1	0	0	0	0
<i>vennomus</i>	0	0	0	0	1	0	0	×	0	0	0	0	+2	0	0	0	0	0	0	0	0				
<i>chaar</i>	0	0	1	0	0	1	1	×	0	0	0	0	0	0	1	-1	-1								
<i>bootet</i>	0	0	0	1	0	0	0	×	0	0	+2	0	0	0	0	0	0								
<i>snaak</i>	0	1	0	0	1	0	0	×	0	0	0	1	-1	-1	0	0	0								
<i>brooken</i>	0	0	0	0	0	1	1	×	0	0	0	0	0	0	0	0	+2								

An attention map

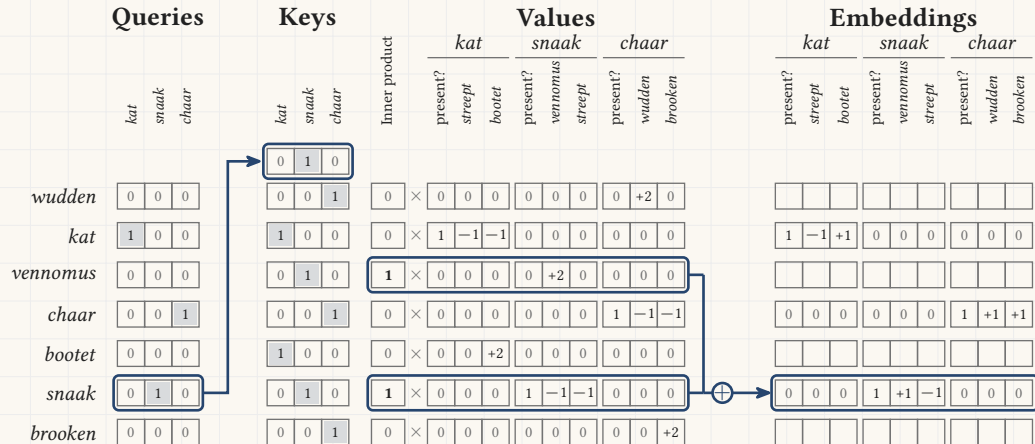


An attention map

Queries			Keys			Values			Embeddings		
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>		<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
									<i>present?</i>	<i>streept</i>	<i>bootet</i>
<i>wudden</i>	0	0	0		0	0	1	0 × 0 0 0	0	0	0
<i>kat</i>	1	0	0		1	0	0	0 × 1 -1 -1	0	0	0
<i>vennomus</i>	0	0	0		0	1	0	1 × 0 0 0	0	+2	0
<i>chaar</i>	0	0	1		0	0	1	0 × 0 0 0	0	0	0
<i>bootet</i>	0	0	0		1	0	0	0 × 0 0 +2	0	0	0
<i>snaak</i>	0	1	0		0	1	0	1 × 0 0 0	1	-1	-1
<i>brooken</i>	0	0	0		0	0	1	0 × 0 0 0	0	0	+2

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
	<i>present?</i>	<i>streept</i>	<i>bootet</i>
<i>wudden</i>			
<i>kat</i>	1	-1	+1
<i>vennomus</i>			
<i>chaar</i>	0	0	0
<i>bootet</i>			
<i>snaak</i>			
<i>brooken</i>			

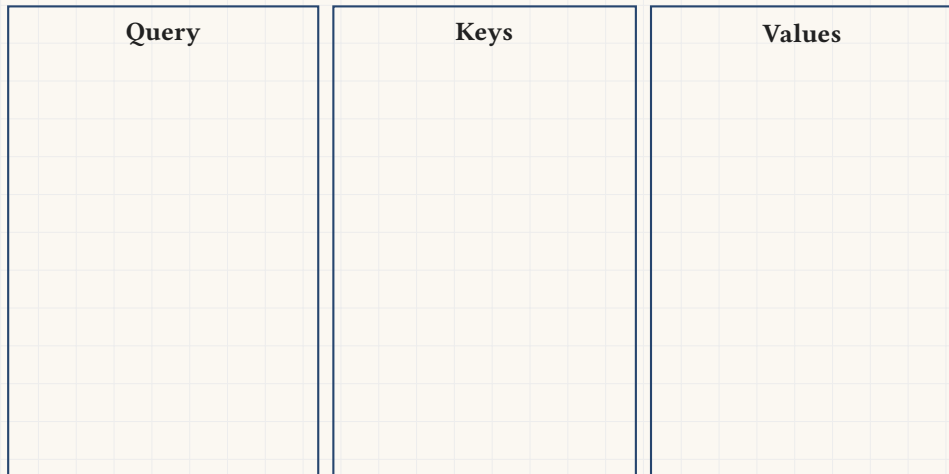
An attention map



An attention map

	Queries	Keys	Values									Embeddings														
	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i>			<i>snaak</i>			<i>chaar</i>			<i>kat</i>			<i>snaak</i>			<i>chaar</i>								
			present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>	present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>						
<i>wudden</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	0	0	0	0	0	0	0	+2	0	0	0	0	0	0	0	0	0	0
0	0	0																								
0	0	1																								
<i>kat</i>	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	1	-1	-1	0	0	0	0	0	0	1	-1	+1	0	0	0	0	0	0
1	0	0																								
1	0	0																								
<i>vennomus</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	0	0	0	0	+2	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0																								
0	1	0																								
<i>chaar</i>	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	0	0	0	0	0	0	1	-1	-1	0	0	0	0	0	0	1	+1	+1
0	0	1																								
0	0	1																								
<i>bootet</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	0	0	+2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0																								
1	0	0																								
<i>snaak</i>	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	0	0	0	1	-1	-1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0																								
0	1	0																								
<i>brooken</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	0	0	0	0	0	0	0	0	+2	0	0	0	0	0	0	0	0	0
0	0	0																								
0	0	1																								

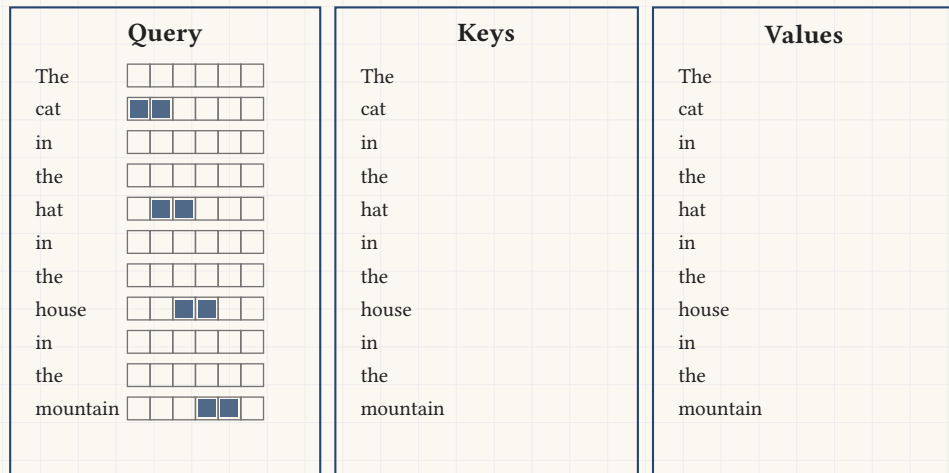
An attention map for location of objects



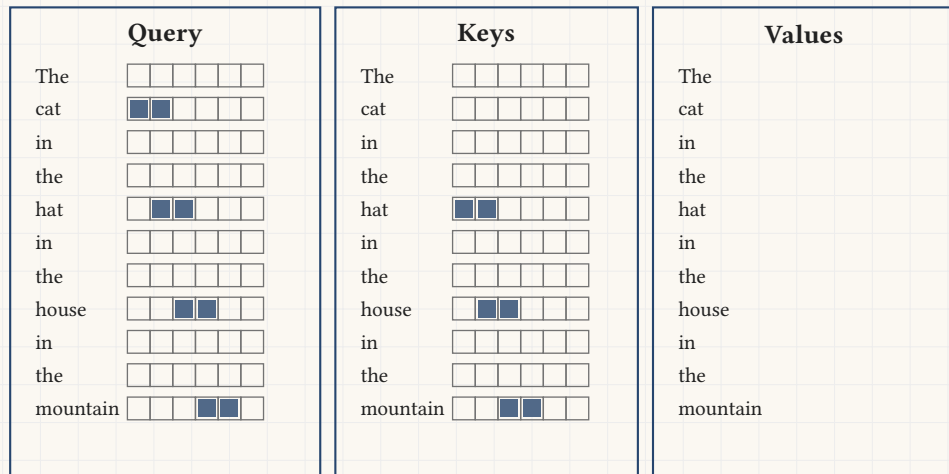
An attention map for location of objects

Query	Keys	Values
The cat in the hat in the house in the mountain	The cat in the hat in the house in the mountain	The cat in the hat in the house in the mountain

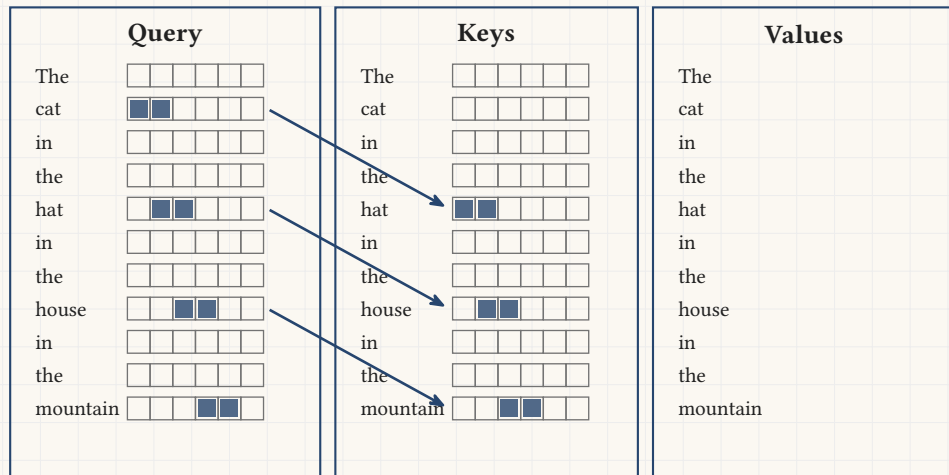
An attention map for location of objects



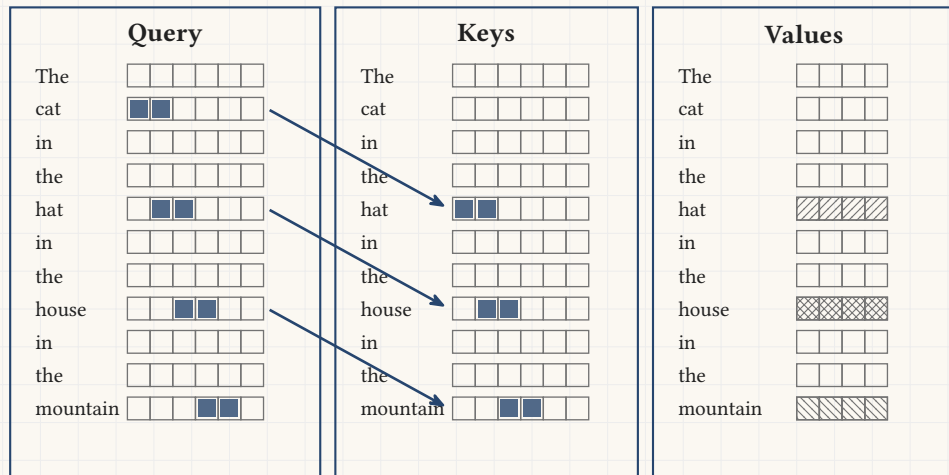
An attention map for location of objects



An attention map for location of objects



An attention map for location of objects

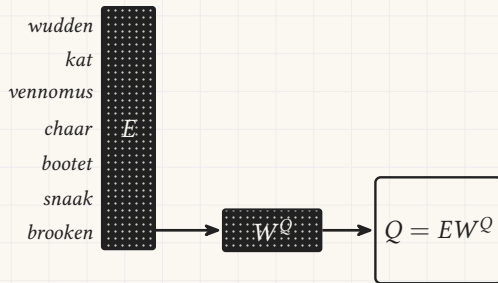


Abstraction of an attention mechanism

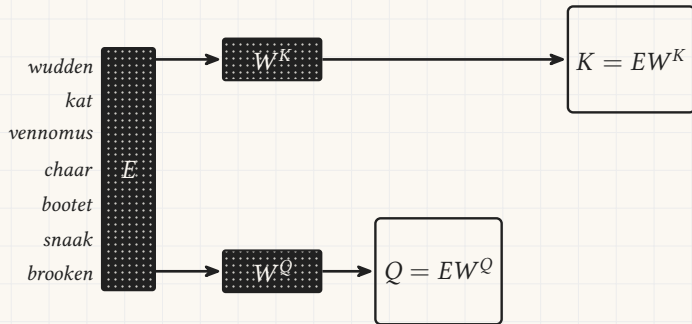
wudden
kat
vennomus
chaar *E*
bootet
snaak
brooken



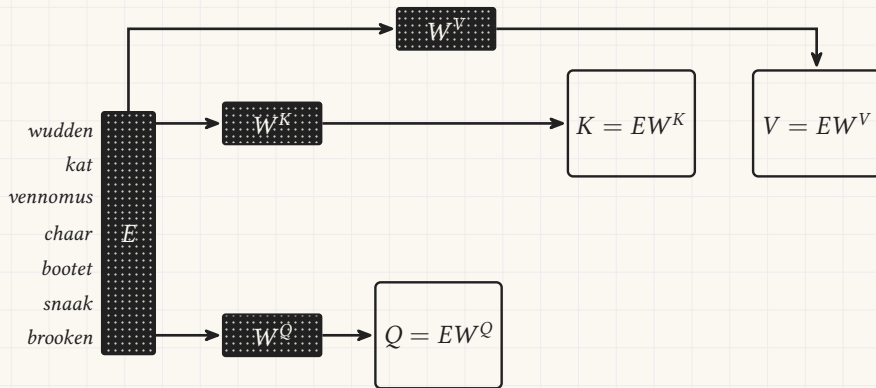
Abstraction of an attention mechanism



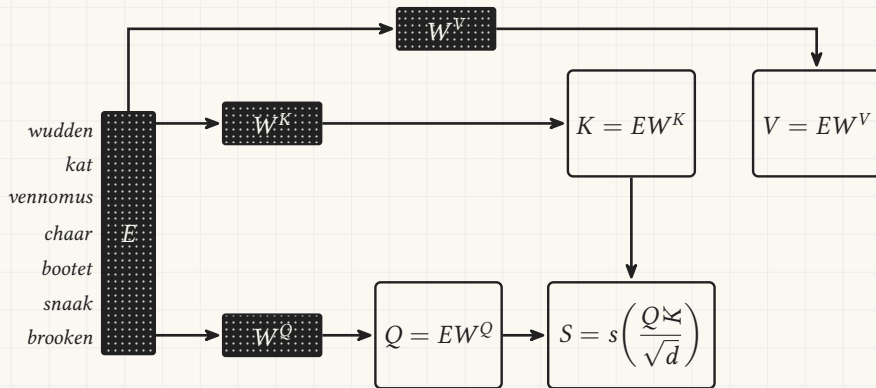
Abstraction of an attention mechanism



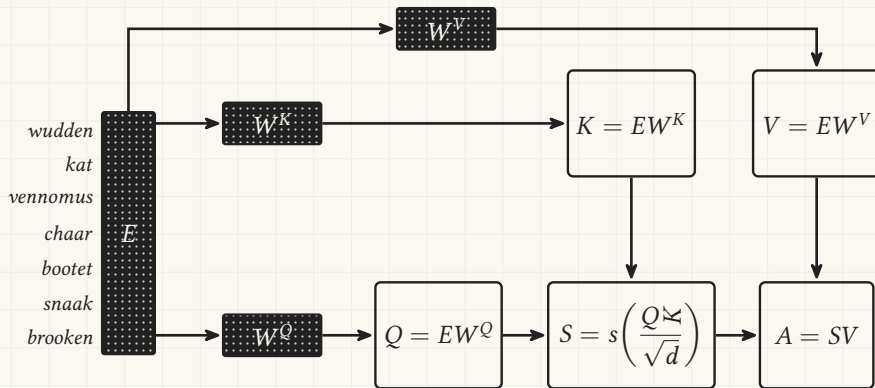
Abstraction of an attention mechanism



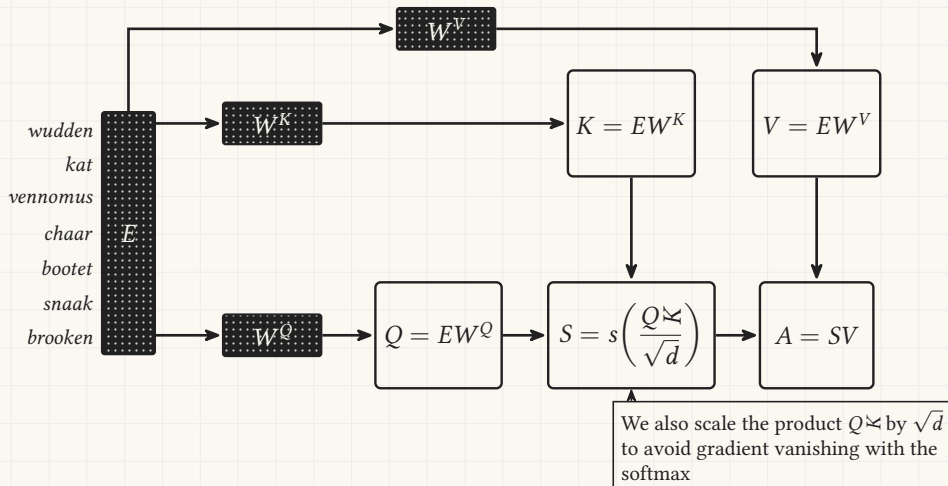
Abstraction of an attention mechanism



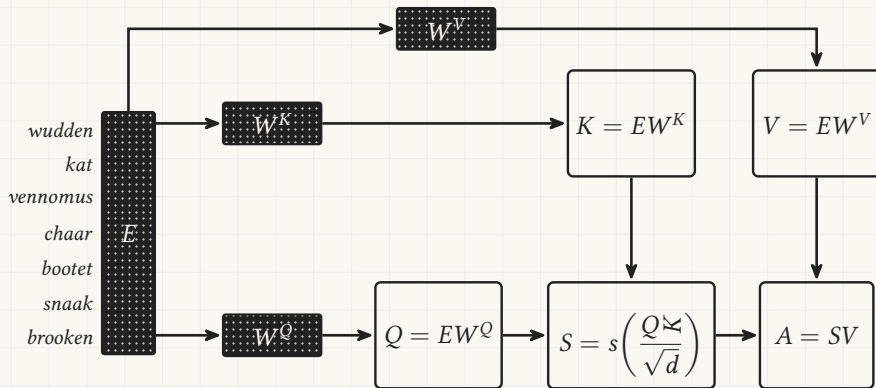
Abstraction of an attention mechanism



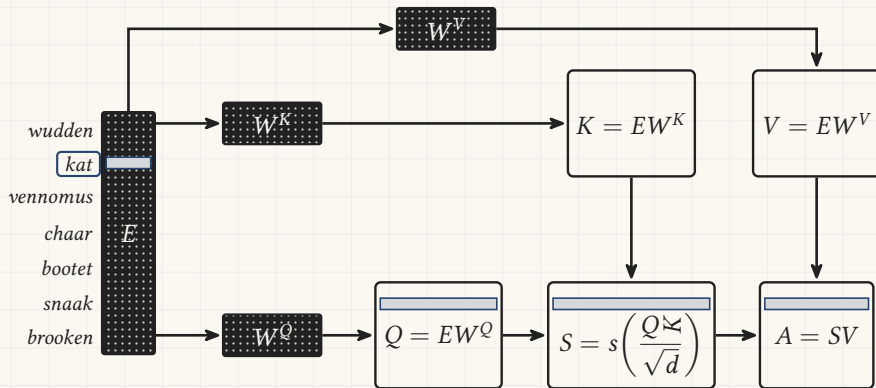
Abstraction of an attention mechanism



Attention from the perspective of information retrieval



Attention from the perspective of information retrieval



Attention from the perspective of information retrieval

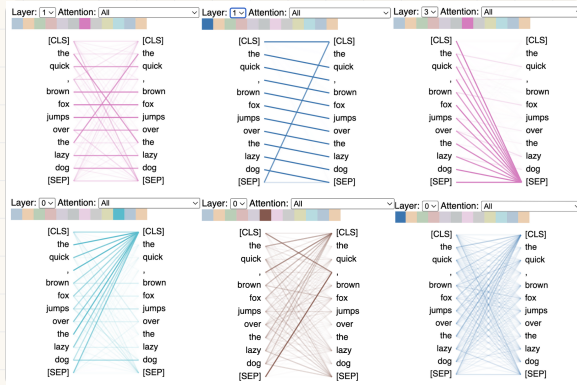


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